

Prompt for Research Assistant (Draft Writing)

Refactoring the Phase II Model into a Progressive Narrative

Gen Li (Team SoloGen)

November 6, 2025

TO: Research Assistant (Draft Writing)
FROM: Gen Li (Team SoloGen)
SUBJECT: Refactoring `p2_bi_model_ggdp.tex` into a Progressive Narrative for Phase II Documentation and Presentation

1 Objective

Your task is to take the final, comprehensive mathematical model described in our draft file `p2_bi_model_ggdp.tex` and rewrite its description.

Instead of presenting it as one final, monolithic model, you must restructure the narrative to show a **logical, three-stage development process**. This narrative will be the backbone of our final report and presentation, demonstrating a rigorous, step-by-step approach to balancing model accuracy with computational feasibility, as required by the competition [?].

We must tell a "story" of how we built our solution.

2 Core Narrative & Guiding Philosophy

Our core philosophy is driven by the competition's "black box" evaluation (the "ORC Battery Degradation Model") and the severe computational demand of "meta-optimization" (finding the optimal 10-year ROI).

Our story is: We didn't just *guess* the right model. We scientifically *built up* from a simple baseline to a "gold standard" academic model. We analyzed the trade-offs at each step and ultimately selected the model that provides the **best practical balance of speed, accuracy, and profitability** for the fixed 10-year horizon.

3 The Three-Stage Narrative (Actionable Plan)

Please structure the new document (and our presentation slides) around these three progressive models. The way of introducing models (i)-(iii) in [?] can be a good reference for you to write our model progression narrative.

3.1 Model (i): The Baseline (The Phase I Model) + aFRR energy Markets part

- **Description:** Start by translating the markdown syntax of the Phase I model described in `whole_project_description.md` [?] into LaTeX syntax, while considering the addition of aFRR energy market dynamics described in the LaTeX file `p2_bi_model_ggdp.tex`. This will form the basis of Model (i).

- **Core Function:** It co-optimizes Day-Ahead, and aFRR energy markets, as well as FCR, and aFRR Capacity markets.
- **Key Flaw to Highlight:** Its *only* degradation-aware component was the rigid **"Daily Cycle Limit"** constraint (our old Cst-5).
- **Punchline:** This is a "naive" approach. It's economically sub-optimal because it treats all cycles (deep vs. shallow) as equally costly and ignores calendar aging entirely.

3.2 Model (ii): The "Practical" Upgrade (Model (i) + Xu-Cyclic)

- **Description:** This is our first major Phase II contribution. We replace the rigid "Daily Cycle Limit" with a flexible, *economic* cost function: C^{cyc} , and also integrate the aFRR energy market dynamics. The approach to do this part is basically take the LaTeX file `p2_bi_model_ggdp.tex` and extract the relevant parts that describe Model (ii) only (i.e., the Phase II model described there without calendar aging).
- **Methodology:** Explain that this model is based on the **Xu et al. (2017) methodology** [?].
- **Key Insights to Emphasize (The "Magic" of this model):**
 1. **"Virtual Segments":** We divide the BESS capacity *virtually* into J segments $e_{\text{soc},j}(t)$, each with a distinct marginal cost c_j^{cost} .
 2. **LP Core:** This model's *aging logic* is a **pure LP**. The optimizer, by simply minimizing cost, will *automatically* use the "cheapest" (shallowest) segments j first.
 3. **The Result:** It perfectly mimics the non-linear "Rainflow" (DOC) cost (deeper cycles = exponentially more expensive) **without** the computational burden of SOS2 constraints.
 4. **Practicality:** It directly tackles our highest (P1) priorities (DOC, C-rate, SoC Range) and its data (a simple DOC vs. Cycle-life curve) is relatively easier to obtain.
- **Punchline:** This model is our "fast, practical workhorse." It captures the most significant (P1) degradation factors while remaining computationally lightweight (an LP-core MILP).

3.3 Model (iii): The "Academic" Upgrade (Model (ii) + Calendar Aging)

- **Description:** Here, we push for maximum academic accuracy by adding the P2 priority: **Calendar Aging** (C^{cal}).
- **Branch (iii-a): The "Gold Standard" (Collath SOS2 Method)**
 - **Methodology:** This is the model *currently described* in our `p2_bi_model_ggdp.tex`. Explain that we add the C^{cal} term based on Collath et al. (2023) [?].
 - **Key Insight (The Cost):** This introduces the **SOS2 constraint**. This is the "gold standard" for accuracy, but it comes at a **severe computational cost**.
 - **The Trade-off:** We must explicitly state this trade-off: "While Model (iii-a) is the most accurate in theory, its reliance on SOS2 transforms our fast model into a complex, slow MILP. This severely hinders our ability to perform the crucial 'meta-optimization' (the α -sweep)."
 - Also, highlight the "epic" data requirement (the multi-dimensional (T, SoC, Q^{loss}) lookup table).
- **Branch (iii-b): The "Engineer's Compromise" (QP Method)**
 - **Methodology:** As a pragmatic alternative to (iii-a), we propose a simplified penalty.

- **Example:** A Quadratic Programming (QP) cost: $C^{\text{cal}} = \alpha_{\text{cal}} \cdot (e_{\text{soc}}(t) - \text{SOC}_{\text{target}})^2$.
- **The Trade-off:** "This model (iii-b) still correctly penalizes high SOC, but avoids the SOS2 bottleneck. It is a robust engineering compromise, faster than (iii-a) but slower than (ii)."

4 The "Meta-Algorithm": The Real Solution Framework

The documentation *must* explain that the "model" (i, ii, or iii) is just the "engine." The "car" that actually wins the race is our **Meta-Algorithm Framework** (from `p2_bi_model_ggdp.tex`, Sections 4.1 & 4.2).

5 Implementation Strategy: A Two-Stage Meta-Optimization

The mathematical formulation defines our "inner-loop" solver. However, the competition's objectives—maximizing a 10-Year ROI by balancing revenue (30%) and degradation (30%)—present a bi-objective problem that is computationally infeasible to solve naively.

5.1 The Computational Challenge: Infeasibility of Naive Meta-Optimization

The core task is to solve the bi-objective function [?]:

$$\max \mathbb{P} - \alpha \cdot C^{\text{Degradation}}$$

Finding the optimal trade-off parameter, α^* , requires a "meta-optimization" loop (i.e., an α -sweep) [?].

A naive implementation would run a full 365-day Rolling Horizon (MPC) simulation for *each* α value. As our analysis shows, this is computationally impossible:

- A 4-hour execution step ($T_{\text{exec}} = 4h$) requires $(365 \text{ days} \times 6 \text{ rolls/day}) = 2190$ MILP solves.
- Assuming a modest 30 seconds per solve, the total time for *one* α value is $2190 \times 30s \approx 1095$ minutes, or **over 18 hours**.
- This makes an α -sweep (e.g., 20+ values) across 5 countries intractable.

5.2 The Two-Stage Solution: Sampled α -Sweep and Full-Horizon Run

To solve this, we decouple the "search" for α^* from the "final execution" of the model.

5.2.1 Stage 1: Sample-Based Meta-Optimization (Finding α^*)

Instead of running the full 365-day simulation for every α , we find α^* using a computationally "cheap" proxy.

1. **Select Sample Data:** We select a small number of "representative" periods (e.g., 3-4 individual weeks) from the 2024 dataset that capture high volatility (winter), low volatility (summer), and average conditions.
2. **Run Fast α -Sweep:** We execute our MPC meta-optimization loop (sweeping 20-30 α values) *only* on these sample weeks.
3. **Extrapolate & Select:** We extrapolate the (Profit vs. Degradation) results from these samples to a full year, calculate the 10-Year ROI for each α , and select the α^* that yields the highest *extrapolated* ROI.

This "sample-based" search can be completed within our 2-hour computational budget.

5.2.2 Stage 2: Full-Year Rolling Horizon (Final Run)

Once the *single* optimal α^* is identified in Stage 1, we perform *one* high-fidelity, full 365-day MPC simulation to generate our final submission files. For this single run, we optimize the MPC window parameters for accuracy and speed:

- **Execution Step (T_{exec}): 24 hours (1 roll per day).**
 - **Justification:** This is our primary time-saving measure. It reduces the total MILP solves from ~ 2190 (for a 4h-roll) to just **365 solves**. Given that the competition provides deterministic data (perfect foresight), the main benefit of a shorter rolling window—managing uncertainty—is not applicable.
- **Planning Horizon ($T_{horizon}$): 48 hours (2-day lookahead).**
 - **Justification:** A 24-hour horizon is "myopic" and will make poor decisions at the end of the day (e.g., failing to prepare for the next morning's price changes). A 48-hour horizon ensures the model can see at least one full 24-hour cycle ahead, allowing it to optimize across daily market cycles effectively.

This two-stage strategy allows us to find the optimal profit-degradation trade-off (α^*) in a feasible timeframe (Stage 1) and then generate an accurate, high-quality result using that parameter (Stage 2).

5.3 Solving the Bi-Objective Problem: The α -Sweep

- Explain that the 30%/30% (Revenue/Degradation) weighting makes this a bi-objective problem [?].
- Our solution is the **Weighted Sum Method**: $\max \left(\text{Profit} - \alpha \cdot (\text{Degradation Cost}) \right)$.
- The "magic number" α is our tuning knob. We find the *best* α^* by running a **meta-optimization loop**: We run the *entire* 365-day MPC simulation for many different α values and pick the one that gives the **maximum 10-Year ROI**.

6 Final Narrative & Conclusion for the Report

Please conclude the documentation with this strategic narrative:

"Our development was a quest for the optimal *practical* trade-off.

- **Model (i)** was too simple.
- **Model (iii-a) (Xu+Collath)** is academically perfect but computationally prohibitive, making the required α -sweep (and thus, finding the true 10-Year ROI) infeasible within the competition's time limits.
- **Model (ii) (Xu-Only)** provides the "sweet spot." It captures the *dominant* P1 degradation factors (DOC/C-rate) with a high-speed, LP-core logic. This speed is a *strategic advantage*, as it allows us to run a far more granular α -sweep (e.g., 50 α -values vs. 5) to find the *true* 10-Year optimal balance.

Given the "black box" nature of the ORC evaluation, a fast, robust, and 'directionally-correct' model (Model ii) that can be optimized *perfectly* (via the α -sweep) is superior to a 'theoretically-perfect' model (Model iii-a) that is too slow to optimize at all."

7 Action Items

- ☐ Create a new LaTeX document for the final report in the same document formatting as the LaTeX file `p2_bi_model_ggdp.tex`.
- ☐ Use `p2_bi_model_ggdp.tex` as the source for the math equations.
- ☐ Restructure the narrative according to the 3-Stage plan above.
- ☐ Integrate the "Meta-Algorithm" (MPC + α -sweep) as the overarching solution framework.
- ☐ Emphasize all "Key Insights" and "Trade-offs" described above.

References

- [1] B. Xu, J. Zhao, T. Zheng, E. Litvinov, and D. S. Kirschen. (2017). "Factoring the Cycle Aging Cost of Batteries Participating in Electricity Markets." *arXiv:1707.04567v2*.
- [2] N. Collath, M. Cornejo, V. Engwerth, H. Hesse, and A. Jossen. (2023). "Increasing the lifetime profitability of battery energy storage systems through aging aware operation." *Applied Energy*, 348, 121531.
- [3] Gen Li. (2025). "Huawei TechArena 2025: BESS Energy Management System." *Internal Project Document*.